Divvy Bike Share Analysis – January 2023

1. Objective

The objective of this analysis is to explore the Divvy bike share data for January 2023 and answer key questions about rider behaviour. Specifically, the project examines who takes more rides, which days have the highest activity, and which bike types are most commonly used.

2. Dataset

- The dataset contains ~190,000 bike rides recorded in January 2023.
- Key columns used in this analysis:
  - ✓ ride_id
  - ✓ member_casual
  - ✓ rideable_type
  - ✓ started_at

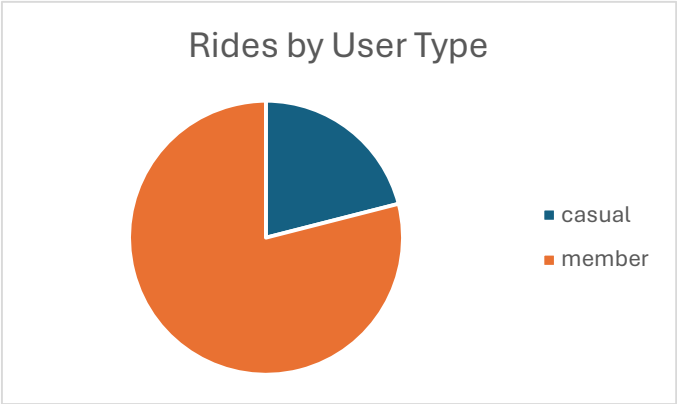
3. Data Cleaning

The dataset was cleaned before analysis.

- Cleaning steps:
  - Checked and removed rows if there were any missing values in key columns (ride_id, member_casual, rideable_type, started_at)
  - Ignored missing values in station names and other columns that were not relevant
  - Converted started_at to proper date format
  - Added a new column for day of the week

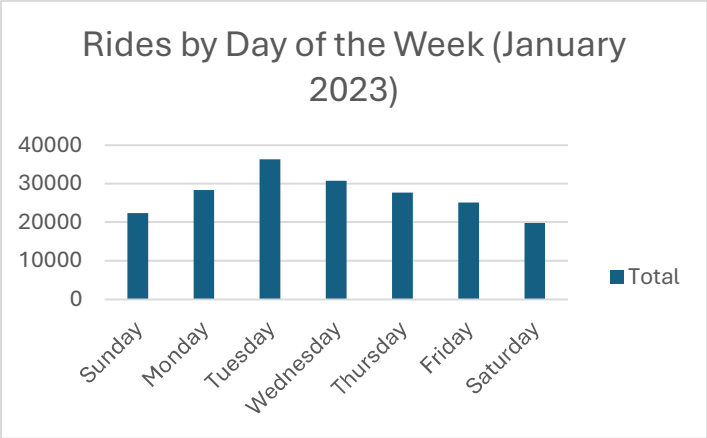
4. Analysis and Visualizations

4.1. Who Takes More Rides?



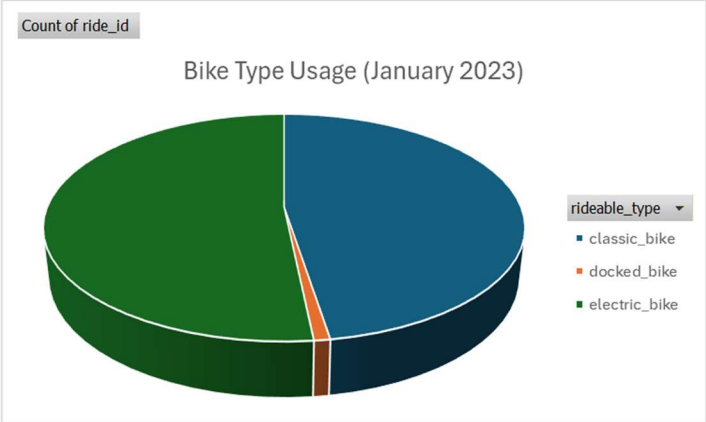
Members take more rides than casual riders, indicating that members use the service more frequently, likely for commuting or regular travel.

4.2. Which Day Has the Most Rides?



Ride activity is highest on Tuesdays, followed by other weekdays. This indicates increased usage on weekdays. This suggests that bike share service is commonly used for weekday commuting rather than only for leisure purposes.

4.3. Which Bike Type is Used Most?



Electric bikes are the most frequently used, indicating that riders prefer a more convenient and less physically demanding option compared to classic or docked bikes.

5. Key Insights

- Members are the primary users of Divvy Bikes
- Ride activity peaks on Tuesdays, suggesting strong weekday usage
- Electric bikes are the most popular choice among riders
- Overall, the service appears to support regular commuting patterns

6. Conclusion

This analysis highlights key usage patterns in Divvy bike share data for January 2023. Members contribute to the majority of rides, indicating frequent and consistent usage. Bike usage peaks on weekdays, particularly on Tuesdays, suggesting strong commuting behaviour. Additionally, the preference for electric bikes suggests that convenience is an important factor for riders. Overall, these findings suggest that the service is primarily used for practical, day-to-day transportation.

7. Recommendations

Based on the analysis, the following recommendations can be made:

- Increase availability of electric bikes during peak days such as Tuesdays
- Ensure sufficient bike supply during weekday commuting hours
- Encourage casual users to convert to memberships through promotions